

More seeds, datasets and runs

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Repository: <https://github.com/myavarih/CLIP-LoRA>

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1. Summary of Evaluated Runs

TABLE 1. Architectural and algorithmic specification of the five evaluated fine-tuning paradigms.

Run	Paradigm Name	Vision Adapter	Text Adapter	Prompt & Class Formulation	Loss Objective
Run 1	Infix CoOp + Ordinal	LoRA ($r=2, \alpha=1.0$)	Frozen Backbone	Infix context ($M=4$) + Learnable class tokens Δe_k	$\mathcal{L}_{ce} + \lambda_{ord}\mathcal{L}_{ord}$
Run 2	Plain Dual LoRA	LoRA ($r=2, \alpha=1.0$)	LoRA ($r=2, \alpha=1.0$)	Static natural template, frozen BPE token embeddings	Standard $\mathcal{L}_{ce}$
Run 3	CoOp-CSC Dual LoRA	LoRA ($r=2, \alpha=1.0$)	LoRA ($r=2, \alpha=1.0$)	Class-Specific Context ($K \times M$ continuous prompt vectors)	Standard $\mathcal{L}_{ce}$
Run 4	Plain LoRA + ResCls	LoRA ($r=2, \alpha=1.0$)	LoRA ($r=2, \alpha=1.0$)	Static natural template + Learnable residual class offset δ_k	Standard $\mathcal{L}_{ce}$
Run 5	MLLM Feat-LoRA	LoRA ($r=2, \alpha=1.0$)	Frozen Backbone	Fine-grained MLLM attribute prompt ensemble + feature residual Δf_k	$\mathcal{L}_{ce} + \lambda_{kg}\mathcal{L}_{kg}$

Note: I forgot to set the flag for Run 3 to use ordinal loss despite the plan. Run 5 implements UniFGVC-inspired MLLM prompt ensembles with feature residual adaptation and kgCoOp anchor regularization ($\lambda_{kg} = 2.0$).

TABLE 2. Benchmark datasets and experimental evaluation suite matrix (216 initial runs on seeds 11, 12, 13, plus 180 runs on seeds 21, 22, 23 with fixed data augmentation; 396 evaluated runs total).

Dataset	Classes (K)	Zero-Shot	Shots (k)	Seeds
Walnut (Initial Aug.)	6	22.26%	{1, 2, 4, 8, 16, 32}	11, 12, 13
Walnut (Fixed Aug.) [*]	6	22.26% [†]	{1, 2, 4, 8, 16, 32}	21, 22, 23
Piarom (Initial Aug.)	5	24.91%	{1, 2, 4, 8, 16, 32}	11, 12, 13
Piarom (Fixed Aug.) [*]	5	24.91% [†]	{1, 2, 4, 8, 16, 32}	21, 22, 23
Cars	196	65.54%	{1, 2, 4, 8, 16, 32}	11, 12, 13

^{*}*Note: Data augmentation was changed (fixed) in the 21, 22, 23 seed runs (letterbox square padding to eliminate object cropping, vertical flipping, subtle translation affine shift, and impulse/Gaussian noise). Runs 1–4 were evaluated across all benchmark suites; Run 5 was evaluated on the fixed-augmentation Walnut and Piarom suites.*

[†]*Standard template zero-shot CLIP baseline is 22.26% (Walnut) and 24.91% (Piarom); the MLLM prompt ensemble zero-shot baseline (Variant 1) is 22.03% (Walnut) and 19.33% (Piarom).*

2. Multi-Dataset Benchmark Results

2.1. Cross-Dataset Performance Trajectory

Figure 1 illustrates the test accuracy progression across shot counts ($k \in \{1, 2, 4, 8, 16, 32\}$) across benchmarks, including the newly evaluated Piarom Date Shape and Walnut Quality Grading suites with fixed non-cropping augmentations and Run 5 (MLLM Feat-LoRA).

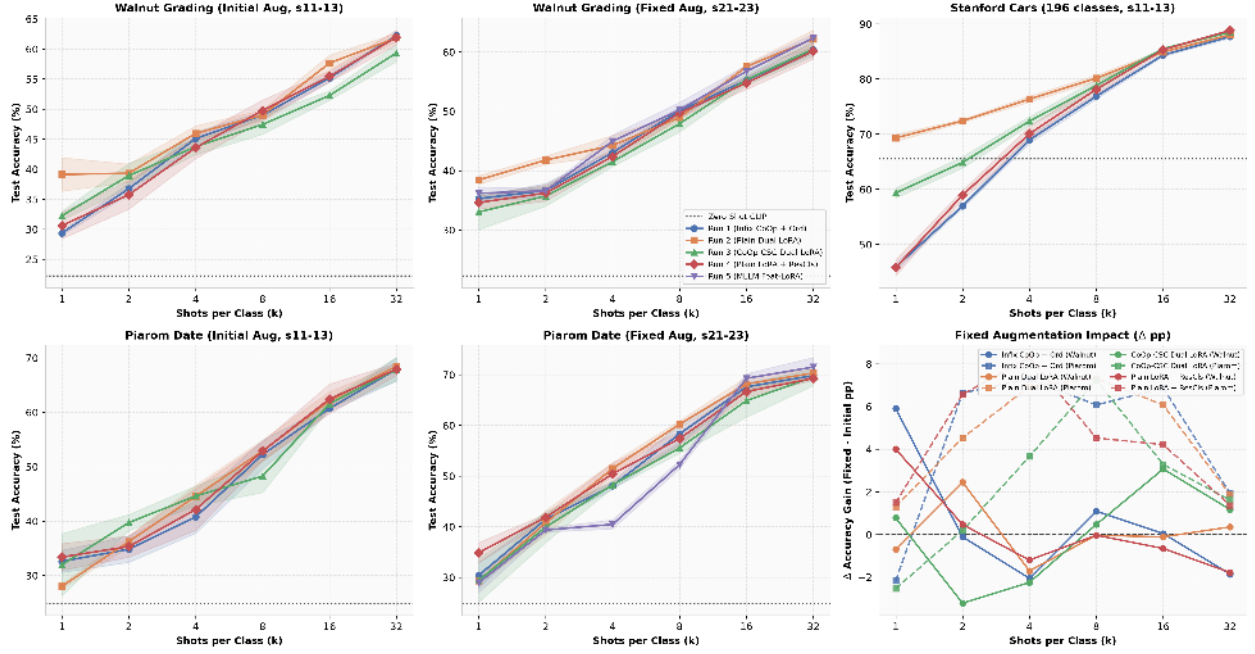


FIGURE 1. Few-shot test accuracy trajectories across 1 to 32 shots on Walnut under initial and fixed data augmentations ($K = 6$), Piarom Date Shape under initial and fixed data augmentations ($K = 5$), and Stanford Cars ($K = 196$). Shaded bands represent ± 1 standard deviation. The dashed line marks the frozen Zero-Shot CLIP baseline.

2.2. The Numbers!

Tables 3, 4, 5, 6, and 7 record the exact numerical test accuracy (mean $\pm$ standard deviation across seeds) for each paradigm and shot level. The top-performing method at each shot is highlighted in green.

TABLE 3. **Walnut Quality Grading — Initial Augmentation** ($K = 6$ classes, Seeds 11, 12, 13). Zero-Shot CLIP Baseline: **22.26%**.

Shots	Run 1 (Infix CoOp+Ord)	Run 2 (Plain LoRA)	Run 3 (CoOp-CSC Dual)	Run 4 (Plain LoRA+ResCls)
1 shots	29.37 $\pm$ 0.35	39.12 $\pm$ 2.79	32.25 $\pm$ 0.76	30.61 $\pm$ 2.20
2 shots	36.75 $\pm$ 0.93	39.32 $\pm$ 1.54	38.93 $\pm$ 2.03	35.74 $\pm$ 2.39
4 shots	45.10 $\pm$ 1.11	45.96 $\pm$ 1.25	43.75 $\pm$ 1.35	43.59 $\pm$ 1.83
8 shots	48.99 $\pm$ 0.55	48.99 $\pm$ 0.84	47.44 $\pm$ 1.63	49.77 $\pm$ 1.75
16 shots	55.17 $\pm$ 0.68	57.65 $\pm$ 1.34	52.25 $\pm$ 0.78	55.44 $\pm$ 0.94
32 shots	62.24 $\pm$ 0.12	61.85 $\pm$ 0.76	59.32 $\pm$ 1.43	61.93 $\pm$ 1.21

TABLE 4. **Walnut Quality Grading — Fixed Data Augmentation** ($K = 6$ classes, Seeds 21, 22, 23). Zero-Shot CLIP Baseline: **22.26%** (MLLM Prompt Ensemble Zero-Shot: **22.03%**).

Shots	Run 1 Infix CoOp+Ord	Run 2 Plain LoRA	Run 3 CoOp-CSC Dual	Run 4 Plain LoRA+ResCls	Run 5 MLLM Feat-LoRA
1 shots	35.28 $\pm$ 0.41	38.42 $\pm$ 0.75	33.02 $\pm$ 2.98	34.62 $\pm$ 1.11	36.17 $\pm$ 1.08
2 shots	36.64 $\pm$ 1.20	41.76 $\pm$ 0.70	35.70 $\pm$ 1.71	36.21 $\pm$ 1.38	36.60 $\pm$ 0.35
4 shots	43.05 $\pm$ 0.49	44.25 $\pm$ 1.69	41.49 $\pm$ 0.82	42.39 $\pm$ 1.06	44.95 $\pm$ 1.08
8 shots	50.08 $\pm$ 0.68	48.95 $\pm$ 0.62	47.90 $\pm$ 1.40	49.73 $\pm$ 0.79	50.23 $\pm$ 1.26
16 shots	55.21 $\pm$ 0.47	57.54 $\pm$ 0.41	55.32 $\pm$ 0.94	54.78 $\pm$ 1.12	56.72 $\pm$ 1.05
32 shots	60.37 $\pm$ 0.65	62.20 $\pm$ 1.50	60.49 $\pm$ 1.23	60.14 $\pm$ 1.34	62.35 $\pm$ 0.71

Note: The data augmentation pipeline was changed (fixed) in the 21, 22, 23 seed runs (letterbox square padding to eliminate object cropping, random vertical flips, translation shifts, and impulse/Gaussian noise).

TABLE 5. **Piarom Date Shape Grading — Initial Augmentation** ($K = 5$ classes, Seeds 11, 12, 13). Zero-Shot CLIP Baseline: **24.91%**.

Shots	Run 1 (Infix CoOp+Ord)	Run 2 (Plain LoRA)	Run 3 (CoOp-CSC Dual)	Run 4 (Plain LoRA+ResCls)
1 shots	32.59 $\pm$ 2.05	28.00 $\pm$ 0.47	31.97 $\pm$ 5.72	33.40 $\pm$ 2.48
2 shots	34.82 $\pm$ 2.43	36.25 $\pm$ 1.03	39.71 $\pm$ 1.49	35.32 $\pm$ 1.96
4 shots	40.77 $\pm$ 3.02	44.55 $\pm$ 2.05	44.61 $\pm$ 1.83	42.13 $\pm$ 3.86
8 shots	52.23 $\pm$ 2.64	52.91 $\pm$ 1.99	48.27 $\pm$ 3.02	52.85 $\pm$ 1.72
16 shots	60.72 $\pm$ 0.60	62.08 $\pm$ 1.70	61.59 $\pm$ 0.84	62.45 $\pm$ 2.74
32 shots	67.91 $\pm$ 2.14	68.46 $\pm$ 0.39	67.91 $\pm$ 2.18	67.91 $\pm$ 1.14

TABLE 6. **Piarom Date Shape Grading — Fixed Data Augmentation** ($K = 5$ classes, Seeds 21, 22, 23). Zero-Shot CLIP Baseline: **24.91%** (MLLM Prompt Ensemble Zero-Shot: **19.33%**).

Shots	Run 1	Run 2	Run 3	Run 4	Run 5
	Infix CoOp+Ord	Plain LoRA	CoOp-CSC Dual	Plain LoRA+ResCls	MLLM Feat-LoRA
1 shots	30.42 $\pm$ 2.54	29.31 $\pm$ 0.47	29.43 $\pm$ 4.30	34.88 $\pm$ 2.00	28.93 $\pm$ 1.93
2 shots	41.45 $\pm$ 0.85	40.77 $\pm$ 2.45	39.90 $\pm$ 3.02	41.88 $\pm$ 0.60	39.34 $\pm$ 0.39
4 shots	48.08 $\pm$ 0.75	51.49 $\pm$ 1.16	48.27 $\pm$ 0.65	50.43 $\pm$ 2.14	40.46 $\pm$ 0.92
8 shots	58.30 $\pm$ 2.09	60.22 $\pm$ 0.67	55.51 $\pm$ 1.06	57.37 $\pm$ 1.37	52.23 $\pm$ 0.85
16 shots	67.60 $\pm$ 0.84	68.15 $\pm$ 1.41	64.87 $\pm$ 3.27	66.67 $\pm$ 1.14	69.27 $\pm$ 0.94
32 shots	69.83 $\pm$ 0.60	70.32 $\pm$ 0.75	69.52 $\pm$ 1.70	69.27 $\pm$ 0.39	71.50 $\pm$ 1.91

Note: The data augmentation pipeline was changed (fixed) in the 21, 22, 23 seed runs (letterbox square padding to eliminate object cropping, random vertical flips, translation shifts, and impulse/Gaussian noise).

TABLE 7. **Stanford Cars Classification** ($K = 196$ classes). Zero-Shot CLIP Baseline: **65.54%**.

Shots	Run 1 (Infix CoOp+Ord)	Run 2 (Plain LoRA)	Run 3 (CoOp-CSC Dual)	Run 4 (Plain LoRA+ResCls)
1 shots	45.85 $\pm$ 0.63	69.30 $\pm$ 0.65	59.31 $\pm$ 0.93	45.74 $\pm$ 1.46
2 shots	56.93 $\pm$ 0.36	72.39 $\pm$ 0.28	64.85 $\pm$ 1.03	58.91 $\pm$ 0.83
4 shots	68.95 $\pm$ 0.67	76.34 $\pm$ 0.55	72.35 $\pm$ 0.81	70.05 $\pm$ 1.34
8 shots	76.85 $\pm$ 0.51	80.14 $\pm$ 0.56	78.82 $\pm$ 1.03	78.13 $\pm$ 1.11
16 shots	84.35 $\pm$ 0.28	84.89 $\pm$ 0.39	85.50 $\pm$ 0.15	85.33 $\pm$ 0.23
32 shots	87.68 $\pm$ 0.30	88.01 $\pm$ 0.22	88.50 $\pm$ 0.21	88.82 $\pm$ 0.06

2.3. Seed Sensitivity and Variance

Figure 2 shows the standard deviation of test accuracy across random seeds (11, 12, 13 for initial augmentation and Cars; 21, 22, 23 for fixed augmentation suites).

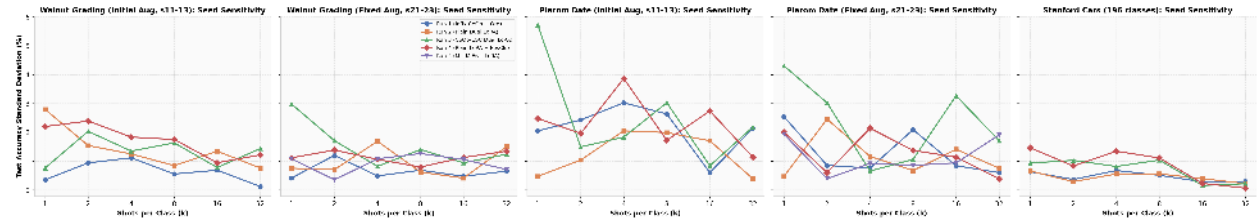


FIGURE 2. Test accuracy standard deviation across random seeds (11, 12, 13 for initial augmentation and Stanford Cars; 21, 22, 23 for fixed augmentation suites).

3. Training Accuracies

Figure 3 and Table 8 report the final training accuracy trajectories and exact numerical values (mean $\pm$ standard deviation across seeds) across shot counts ($k \in \{1, 2, 4, 8, 16, 32\}$) for each evaluated paradigm. The top-performing method at each shot is highlighted in green.

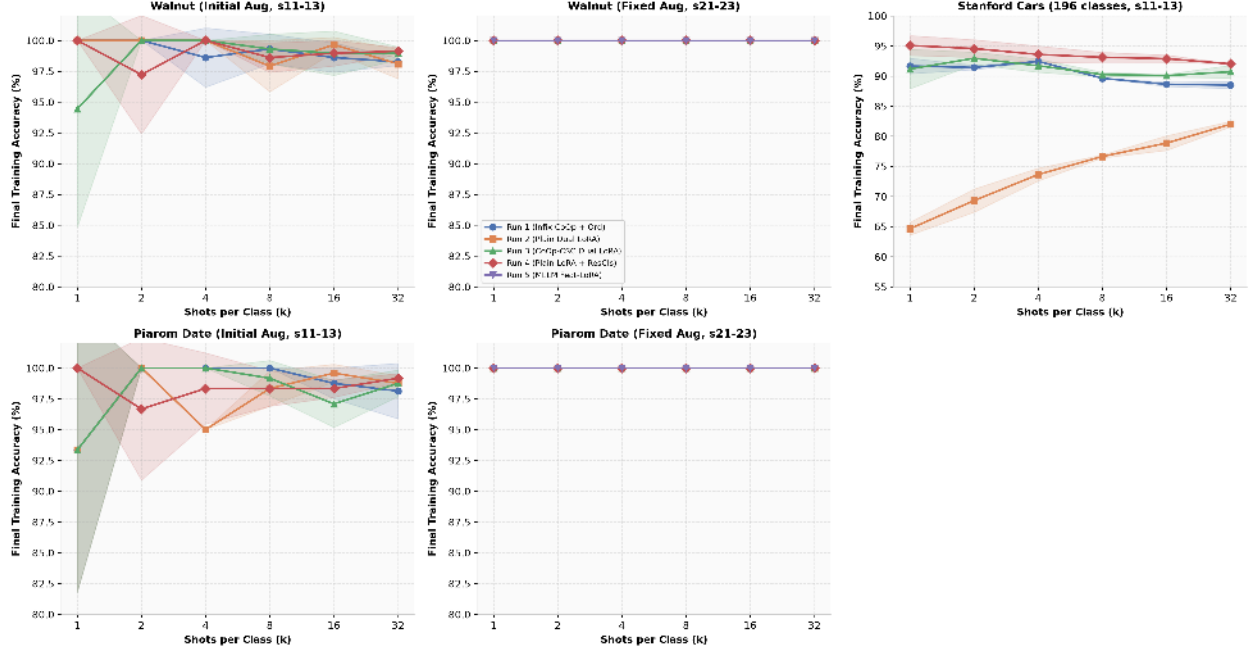


FIGURE 3. Few-shot final training accuracy trajectories across 1 to 32 shots on Walnut under initial and fixed data augmentations ($K = 6$), Pirom Date Shape under initial and fixed data augmentations ($K = 5$), and Stanford Cars ($K = 196$). Shaded bands represent ± 1 standard deviation.

TABLE 8. **Final Training Accuracy Across Benchmarks** (mean $\pm$ standard deviation across seeds).

Shots	Run 1 Infix CoOp+Ord	Run 2 Plain LoRA	Run 3 CoOp-CSC Dual	Run 4 Plain LoRA+ResCls	Run 5 MLLM Feat-LoRA
Walnut Quality Grading — Initial Augmentation ($K = 6$ classes, Seeds 11, 12, 13)					
1 shots	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	94.44 $\pm$ 9.62	100.00 $\pm$ 0.00	—
2 shots	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	97.22 $\pm$ 4.81	—
4 shots	98.61 $\pm$ 2.41	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	—
8 shots	99.31 $\pm$ 1.20	97.92 $\pm$ 2.08	99.31 $\pm$ 1.20	98.61 $\pm$ 1.20	—
16 shots	98.61 $\pm$ 1.20	99.65 $\pm$ 0.60	98.96 $\pm$ 1.80	98.96 $\pm$ 1.04	—
32 shots	98.26 $\pm$ 0.30	98.09 $\pm$ 1.20	98.96 $\pm$ 0.52	99.13 $\pm$ 0.30	—
Walnut Quality Grading — Fixed Augmentation ($K = 6$ classes, Seeds 21, 22, 23)					
1 shots	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00
2 shots	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00
4 shots	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00
8 shots	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00
16 shots	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00
32 shots	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00
Piarom Date Shape Grading — Initial Augmentation ($K = 5$ classes, Seeds 11, 12, 13)					
1 shots	93.33 $\pm$ 11.55	93.33 $\pm$ 11.55	93.33 $\pm$ 11.55	100.00 $\pm$ 0.00	—
2 shots	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	96.67 $\pm$ 5.77	—
4 shots	100.00 $\pm$ 0.00	95.00 $\pm$ 0.00	100.00 $\pm$ 0.00	98.33 $\pm$ 2.89	—
8 shots	100.00 $\pm$ 0.00	98.33 $\pm$ 1.44	99.17 $\pm$ 1.44	98.33 $\pm$ 1.44	—
16 shots	98.75 $\pm$ 1.25	99.58 $\pm$ 0.72	97.08 $\pm$ 1.91	98.33 $\pm$ 0.72	—
32 shots	98.12 $\pm$ 2.25	98.75 $\pm$ 0.62	98.75 $\pm$ 1.08	99.17 $\pm$ 0.36	—
Piarom Date Shape Grading — Fixed Augmentation ($K = 5$ classes, Seeds 21, 22, 23)					
1 shots	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00
2 shots	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00
4 shots	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00
8 shots	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00
16 shots	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00
32 shots	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00
Stanford Cars Classification ($K = 196$ classes, Seeds 11, 12, 13)					
1 shots	91.67 $\pm$ 1.28	64.63 $\pm$ 1.06	91.16 $\pm$ 3.28	95.07 $\pm$ 1.64	—
2 shots	91.41 $\pm$ 0.39	69.30 $\pm$ 1.93	92.94 $\pm$ 1.03	94.56 $\pm$ 1.47	—
4 shots	92.43 $\pm$ 0.07	73.64 $\pm$ 1.06	91.71 $\pm$ 1.11	93.58 $\pm$ 1.40	—
8 shots	89.63 $\pm$ 0.04	76.64 $\pm$ 0.26	90.26 $\pm$ 0.36	93.11 $\pm$ 0.83	—
16 shots	88.63 $\pm$ 0.37	78.86 $\pm$ 1.22	90.06 $\pm$ 0.23	92.88 $\pm$ 0.61	—
32 shots	88.50 $\pm$ 0.61	82.00 $\pm$ 0.46	90.70 $\pm$ 1.08	92.02 $\pm$ 0.02	—